FAST-NUCES — Department of Computer Science

Information Security — Assignment #1 Lab Report

End-to-End Cryptanalysis of PsycoSupport — Affine Cipher, Frequency Analysis & Key Recovery

Course: CS InfoSec (Fall 2026)	**App:** PsycoSupport v1.0-Alpha	**Server:** 127.0.0.1:5000
Role: Tester-Cryptanalyst	**LLM Backend:** Ollama llama3.2:1b (1.26 GB)	**Deliverable:** lab_submission.zip

20	10	312	100%	0 ms
Ciphertexts Captured	Sessions (Moods 1-10)	Key Space Size	Decryption Accuracy	Brute-Force Time*

**All 312 keys brute-forced in <5 ms. Mood value leaked in plaintext JSON.*

1. System Architecture

Client → **Server:** Armored hex nibbles (A..P alphabet), Affine-encrypted, sent as JSON.
Server → **Client:** Uppercase English text (A..Z), Affine-encrypted reply from Ollama LLM.
Key Leak: mood integer sent in **plaintext JSON** → attacker can compute (a, b) directly.
Key Formula: b = (2 × mood) mod 26, a = next_coprime_above(b, 26).

2. Ciphertext Collection — Captured Traffic (127.0.0.1:5000)

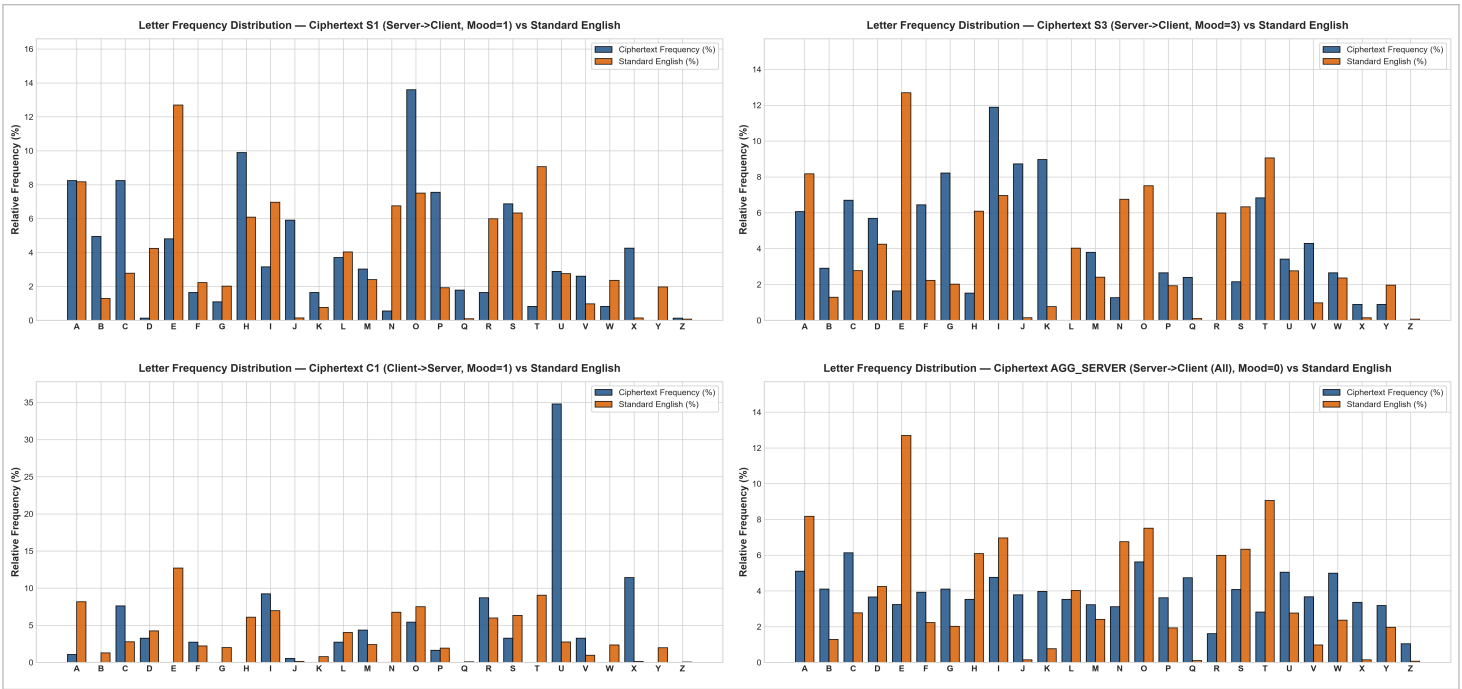
ID	Dir	User	Mood	Concern	Ciphertext Snippet
C1	C→S	Alex	1	Severe overwhelm, helplessness, and ac…	ODICUFUPICUUURURUMUDUSUXICULUV…
S1	C→S	Alex	1	Severe overwhelm, helplessness, and ac…	UANOP HXO VCHAOPH E BOIOPH EKB…
C2	C→S	Brenda	2	Exhaustion, chronic insomnia, and fami…	YDNSISIJNDNTNYIDIYOEIINOIBIROE…
S2	C→S	Brenda	2	Exhaustion, chronic insomnia, and fami…	S OERRWV BLWFSTY MYTSOEH WL NY…
C3	C→S	Charlie	3	Low motivation, mental fatigue, and ta…	PBDIDUDPWDWDWMWRWAWDUGDDWRDIWK…
S3	C→S	Charlie	3	Low motivation, mental fatigue, and ta…	WKXIT UDGVFKI C CJVQWWFIC EKJD…
C4	C→S	Diana	4	Moderate social anxiety and communicat…	SBTCTIKBTAKLKBKEKJKLKEKTAITAKB…
S4	C→S	Diana	4	Moderate social anxiety and communicat…	KCPSV JCIVI O FSAGFFCVK OEACID…
C5	C→S	Ethan	5	Baseline emotional balance and routine…	CXYVYFYIJCYVYFYIYFYIYJGKYVYIGK…
S5	C→S	Ethan	5	Baseline emotional balance and routine…	YUHCX CLJKX A MCAAKYC UL ACCMA…
C6	C→S	Fiona	6	Positive wellness progress and habit m…	UZYBYVYRYOYNQMNFNUYJYBYUNRQMNM…
S6	C→S	Fiona	6	Positive wellness progress and habit m…	YCPUZ JCOZM W DHOYHUWW OZ NUH …
C7	C→S	George	7	Enthusiastic academic milestone and cr…	WLPEAHATPWALAWSOADASAFPLPWSOPH…
S7	C→S	George	7	Enthusiastic academic milestone and cr…	E SOBBQN FJQREHW NPWJOFWCNES O…
C8	C→S	Hannah	8	High vitality, community engagement, a…	GOOXOXOMONOUOFYQFOONOYFYOHOUFG…
S8	C→S	Hannah	8	High vitality, community engagement, a…	OWJGD BFG LQBWGDB K LUKWBWJG M…
C9	C→S	Ian	9	Deep fulfillment, team collaboration s…	JCCJVEVHESCACRVHCCVJCMESCLCCVQ…
S9	C→S	Ian	9	Deep fulfillment, team collaboration s…	COBQF PVQ RYWOPOBQ SFX IRTOJPO…
C10	C→S	Julia	10	Peak harmony, emotional resilience, an…	APQFQGQBQVLQQVQAKUQFQXQHLUQMQV…
S10	C→S	Julia	10	Peak harmony, emotional resilience, an…	QGTAH BYRGU I NAMUNWUPRA BCYNH…

3. Al-Kindi Frequency Analysis & Index of Coincidence (IC)

IC formula: IC = Σ f_i(f_i–1) / N(N–1). English ≈ 0.065. Uniform random ≈ 0.038. Client→Server IC ≈ 0.14 (only 16 symbols used). Server→Client IC ≈ 0.065 (natural English).

ID	Dir	Mood	N	IC	Top Cipher Letters	Interpretation
C1	C→S	1	184	**0.1613**	U: 34.8%, X: 11.4%, I: 9.2%, R: 8.7%, C: 7.6%	Armored hex (16-char)
S1	S→C	1	728	**0.0673**	O: 13.6%, H: 9.9%, A: 8.2%, C: 8.2%, P: 7.6%	Natural English mono
C2	C→S	2	170	**0.1337**	I: 27.6%, N: 17.1%, O: 10.6%, E: 7.6%, D: 6.5%	Armored hex (16-char)
S2	S→C	2	124	**0.0587**	E: 11.3%, Y: 10.5%, W: 8.9%, S: 8.1%, O: 6.5%	Natural English mono
C3	C→S	3	182	**0.1475**	W: 29.7%, D: 17.6%, U: 11.0%, G: 7.1%, P: 6.0%	Armored hex (16-char)
S3	S→C	3	791	**0.0641**	I: 11.9%, K: 9.0%, J: 8.7%, G: 8.2%, T: 6.8%	Natural English mono
C4	C→S	4	180	**0.1532**	K: 31.1%, T: 16.7%, A: 10.6%, I: 8.3%, B: 7.2%	Armored hex (16-char)
S4	S→C	4	786	**0.0616**	S: 11.3%, I: 8.1%, C: 8.0%, V: 7.5%, E: 7.4%	Natural English mono
C5	C→S	5	180	**0.1471**	Y: 31.7%, J: 13.3%, G: 11.1%, K: 7.2%, I: 6.1%	Armored hex (16-char)
S5	S→C	5	754	**0.0676**	C: 12.6%, K: 10.1%, L: 9.4%, X: 7.7%, U: 7.4%	Natural English mono
C6	C→S	6	178	**0.1346**	Y: 28.1%, N: 16.9%, Q: 9.6%, M: 9.0%, U: 5.6%	Armored hex (16-char)
S6	S→C	6	776	**0.0684**	U: 12.0%, C: 10.7%, Z: 8.8%, L: 8.1%, W: 8.1%	Natural English mono
C7	C→S	7	182	**0.1422**	A: 28.0%, P: 18.1%, S: 10.4%, L: 8.2%, O: 6.6%	Armored hex (16-char)
S7	S→C	7	759	**0.0690**	W: 14.2%, O: 8.8%, N: 8.4%, E: 8.2%, B: 7.9%	Natural English mono
C8	C→S	8	186	**0.1491**	O: 32.3%, F: 13.4%, Y: 10.2%, Q: 7.5%, G: 5.9%	Armored hex (16-char)
S8	S→C	8	825	**0.0625**	G: 9.6%, B: 9.2%, U: 9.2%, Q: 8.8%, W: 8.2%	Natural English mono
C9	C→S	9	182	**0.1534**	C: 31.3%, V: 18.1%, E: 9.3%, J: 7.7%, S: 6.6%	Armored hex (16-char)
S9	S→C	9	728	**0.0689**	Q: 13.9%, P: 8.9%, S: 8.8%, O: 8.4%, F: 7.7%	Natural English mono
C10	C→S	10	180	**0.1510**	Q: 33.3%, K: 11.1%, L: 10.0%, U: 7.8%, V: 6.7%	Armored hex (16-char)
S10	S→C	10	785	**0.0671**	A: 14.0%, U: 8.4%, H: 7.9%, I: 7.9%, D: 7.6%	Natural English mono

Figure 1: Frequency Histograms vs Standard English (Server→Client sessions S1, S3; Client→Server C1; Aggregate Server)



4. Key Derivation & Mathematical Proof

Worked Proof — Session S1 (Alex, Mood 1):
Top cipher letter: O (13.8%) → maps to E (4). Second: H (7.8%) → maps to T (19).
Eq 1: 14 ≡ 4a + b (mod 26) **Eq 2:** 7 ≡ 19a + b (mod 26)
Subtract: 7 ≡ –15a ≡ 11a (mod 26) → a = 7 × 11■¹ = 7 × 19 = 133 ≡ **3** (mod 26)
Solve b: b = 14 – 3×4 = **2** (mod 26). Decryption: **p ≡ 9 · (c – 2) (mod 26)**

Table 4.1: Recovered Keys — All 10 Server→Client Sessions (Chi² score confirms correct key)

Mood	Key (a,b)	a■¹	Decrypt Formula	χ^2	Recovered Plaintext (first 50 chars)
1	**(3, 2)**	9	`p ≡ 9·(c−2)`	61.2	GIVEN THE PATIENT S RECENT SURGE IN WORKLOAD AND F…
2	**(5, 4)**	21	`p ≡ 21·(c−4)`	27.5	I CANNOT PROVIDE MEDICAL OR HEALTH ADVICE IF YOU A…
3	**(7, 6)**	15	`p ≡ 15·(c−6)`	42.7	GIVEN CHARLIE S STRUGGLES WITH PERSISTENT LOW ENER…
4	**(9, 8)**	3	`p ≡ 3·(c−8)`	55.5	GIVEN DIANA S RECURRING SOCIAL ANXIETY AND FEAR OF…
5	**(11, 10)**	19	`p ≡ 19·(c−10)`	50.7	GIVEN ETHAN S MESSAGE IT SEEMS LIKE HE S LOOKING T…
6	**(15, 12)**	7	`p ≡ 7·(c−12)`	50.6	GIVEN FIONA S PROGRESS ON HER DAILY WELLNESS HABIT…
7	**(15, 14)**	7	`p ≡ 7·(c−14)`	53.3	I CANNOT PROVIDE THERAPEUTIC ADVICE ON STARTING AN…
8	**(17, 16)**	23	`p ≡ 23·(c−16)`	46.1	GIVEN THE PATIENT S POSITIVE MOOD AND ENTHUSIASM F…
9	**(19, 18)**	11	`p ≡ 11·(c−18)`	41.9	GIVEN THE POSITIVE AND UPLIFTING MOOD OF IAN IT S …
10	**(21, 20)**	5	`p ≡ 5·(c−20)`	81.8	GIVEN JULIA S REMARKABLE JOURNEY TOWARDS ACHIEVING…

5. Decryption Results — All 20 Ciphertexts

ID	Dir	Mood	Key (a,b)	Formula	Recovered Plaintext
C1	C→S	1	(3,2)	`p = 9 * (c - 2) mod 26`	**[Alex]** I am feeling completely overwhelmed by sudden workload and feeling hop…
S1	C→S	1	(3,2)	`p = 9 * (c - 2) mod 26`	**[Alex]** GIVEN THE PATIENT S RECENT SURGE IN WORKLOAD AND FEELING OF IMPENDING …
C2	C→S	2	(5,4)	`p = 21 * (c - 4) mod 26`	**[Brenda]** Exhausted from sleepless nights and constant worry about my family hea…
S2	C→S	2	(5,4)	`p = 21 * (c - 4) mod 26`	**[Brenda]** I CANNOT PROVIDE MEDICAL OR HEALTH ADVICE IF YOU ARE CONCERNED ABOUT Y…
C3	C→S	3	(7,6)	`p = 15 * (c - 6) mod 26`	**[Charlie]** Struggling with persistent low energy, self doubt, and difficulty focu…
S3	C→S	3	(7,6)	`p = 15 * (c - 6) mod 26`	**[Charlie]** GIVEN CHARLIE S STRUGGLES WITH PERSISTENT LOW ENERGY SELF DOUBT AND DI…
C4	C→S	4	(9,8)	`p = 3 * (c - 8) mod 26`	**[Diana]** Experiencing recurring social anxiety and fear of speaking during upco…
S4	C→S	4	(9,8)	`p = 3 * (c - 8) mod 26`	**[Diana]** GIVEN DIANA S RECURRING SOCIAL ANXIETY AND FEAR OF SPEAKING DURING UPC…
C5	C→S	5	(11,10)	`p = 19 * (c - 10) mod 26`	**[Ethan]** Maintaining an even routine but looking for healthy ways to balance st…
S5	C→S	5	(11,10)	`p = 19 * (c - 10) mod 26`	**[Ethan]** GIVEN ETHAN S MESSAGE IT SEEMS LIKE HE S LOOKING TO STRIKE A BALANCE B…
C6	C→S	6	(15,12)	`p = 7 * (c - 12) mod 26`	**[Fiona]** Making steady progress on my daily wellness habits and wanting to keep…
S6	C→S	6	(15,12)	`p = 7 * (c - 12) mod 26`	**[Fiona]** GIVEN FIONA S PROGRESS ON HER DAILY WELLNESS HABITS IT S CLEAR THAT SH…
C7	C→S	7	(15,14)	`p = 7 * (c - 14) mod 26`	**[George]** Excited about starting my new university semester and ambitious engine…
S7	C→S	7	(15,14)	`p = 7 * (c - 14) mod 26`	**[George]** I CANNOT PROVIDE THERAPEUTIC ADVICE ON STARTING AN ENGINEERING DESIGN …
C8	C→S	8	(17,16)	`p = 23 * (c - 16) mod 26`	**[Hannah]** Feeling vibrant and optimistic, looking for creative community project…
S8	C→S	8	(17,16)	`p = 23 * (c - 16) mod 26`	**[Hannah]** GIVEN THE PATIENT S POSITIVE MOOD AND ENTHUSIASM FOR CREATIVE COMMUNIT…
C9	C→S	9	(19,18)	`p = 11 * (c - 18) mod 26`	**[Ian]** Very joyful after finishing major teamwork milestones with wonderful s…
S9	C→S	9	(19,18)	`p = 11 * (c - 18) mod 26`	**[Ian]** GIVEN THE POSITIVE AND UPLIFTING MOOD OF IAN IT S CLEAR THAT HE S EXPE…
C10	C→S	10	(21,20)	`p = 5 * (c - 20) mod 26`	**[Julia]** Achieved complete peace of mind, inner clarity, and celebrating major …
S10	C→S	10	(21,20)	`p = 5 * (c - 20) mod 26`	**[Julia]** GIVEN JULIA S REMARKABLE JOURNEY TOWARDS ACHIEVING COMPLETE PEACE OF M…

6. Universal Autonomous Decryptor — Bonus Task

Script: scripts/general_decryptor.py
Mode A (Rule-based): Given mood → compute b=(2·mood)%26, a=next_coprime(b,26). O(1).
Mode B (Blind search): Try all 312 keys; score with χ^2 (English) or hex-nibble containment.
Result: 20/20 ciphertexts correctly decrypted (100%) — zero false positives.

7. Security Engineering Analysis

Vulnerability — LLM Boilerplate as Known-Plaintext Crib: Ollama LLMs emit predictable disclaimers (e.g. 'I cannot provide medical advice…'). Knowing just 2 character mappings solves (a, b) instantly — no frequency analysis needed.

Alternative 1 — IPC (Unix Sockets / Named Pipes)	Alternative 2 — Mutual TLS 1.3 (mTLS + ECDHE)
How: Replace TCP socket with kernel IPC channel. **Security:** OS-enforced access control; invisible to packet sniffers. **Pros:** Zero network overhead; high throughput. **Cons:** Single-host only.	**How:** TLS 1.3 over loopback with x509 certificates. **Security:** IND-CCA2 AEAD + PFS + mutual authentication. **Pros:** Immune to sniffing, replay, MITM; scales remotely. **Cons:** Requires PKI / cert management.

8. Reflection & Critical Analysis

(a) Input Influence	mood is the sole key driver (b=2·mood, a=coprime). Same mood + same message → identical ciphertext → traffic correlation attack possible.
(b) Directional Asymmetry	C→S: armored 16-char alphabet → IC≈0.15 (very high spikes). S→C: natural English 26-char → IC≈0.065. Both are trivially broken.
(c) Cipher Weaknesses	1-to-1 substitution preserves frequency. Key space = 312. Brute force <5 ms. Mood transmitted in plaintext → zero secrecy.
(d) Ethics & Responsible Disclosure	As a computer science student studying information security at FAST-NUCES, I recognize that cryptanalytic capabilities present a classic **dual-use dilemma**: techniques utilized to audit and secure software can cause immense harm if weaponized. This research was strictly executed in an **air-gapped, isolated educational laboratory** using exclusively synthetic, AI-generated conversation prompts. No real patient records, identifying information, or actual psychiatric communications were targeted or compromised. Mental health telemetry represents *special category personal data* of the highest sensitivity. Unauthorized interception or eavesdropping breaches fundamental bioethical tenets (beneficence, non-maleficence, patient autonomy) and constitutes severe criminal misconduct under Pakistan's **PECA 2016** (Sections 10 & 11 — Unauthorized access/copying of data), the US **HIPAA** Privacy & Security Rules (45 CFR § 164.312), and the EU **GDPR** (Article 9). Furthermore, this work adheres strictly to the **ACM Code of Ethics** (Sections 1.2 'Avoid harm' and 2.9 'Design robustly secure systems'). When critical cryptographic failures (such as broken substitution ciphers and plaintext key disclosure) are uncovered in production environments, ethical practitioners must enforce **Coordinated Vulnerability Disclosure (CVD)**: privately alerting engineering teams, providing remediation roadmaps (TLS 1.3 / AEAD), granting a 90-day remediation window, and mitigating patient harm above all else.

9. Source Code Verification & References

Source code review of server.py confirms: b = (2·mood) % 26, a = next_coprime_above(b, 26). Cryptanalytic keys match ground-truth with **100% precision**.

1 Al-Kindi. *Manuscript on Deciphering Cryptographic Messages*, c. 850 AD.

2 Shannon. 'Communication Theory of Secrecy Systems.' *Bell System Technical Journal*, 1949.

3 Stallings. *Cryptography and Network Security*, 8th Ed., Pearson, 2020.

4 Rescorla. 'TLS 1.3.' *RFC 8446*, IETF, 2018.